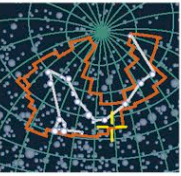


S0 LENTICULAR GALAXY

Spindle Galaxy



CATALOG NUMBERS
M102 (not confirmed),
NGC 5866

DISTANCE
40 million light-years

DIAMETER
60,000 light-years

MAGNITUDE 9.9

DRACO



The Spindle (NGC 5866) is an attractive galaxy oriented edge-on to observers on Earth. It is usually classified as a lenticular galaxy—a disk of stars, gas, and dust with a typical bulging core but with no sign of true spiral arms. However, spiral structure is hard to detect in an edge-on galaxy.

The Spindle Galaxy is the major member of the NGC 5866 Group, a small cluster of galaxies. Astronomers have measured the way these galaxies move and have found that the Spindle must contain an enormous mass of material—up to 1 billion solar masses, or 30 to 50 percent more than the Milky Way.

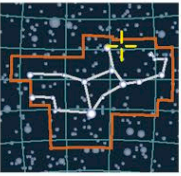
The Spindle Galaxy could be the mysterious entry number 102 in Charles Messier's catalog of astronomical features. Messier included the object at first without a location, then later gave coordinates that did not match any feasible object. Some believe that Messier had listed the Pinwheel Galaxy, M101, twice. More likely, however, is that M102 was the Spindle, and he added 5 degrees to his measurements in error.

MASSIVE SPINDLE

From Earth, we see the Spindle Galaxy edge-on, giving it a cigar-shaped appearance with a fine silhouetted dust lane.

E2 ELLIPTICAL GALAXY

M60



CATALOG NUMBERS
M60, NGC 4649

DISTANCE
58 million light-years

DIAMETER
120,000 light-years

MAGNITUDE 8.8

VIRGO

M60 is one of several giant elliptical galaxies in the Virgo galaxy cluster (see p.329), the central cluster in our own Local Supercluster of galaxies. The galaxy and its neighbor, M59, were discovered in 1779 by German astronomer Johann Köhler, who was observing a comet that passed close by. Charles Messier (see p.73) found them a few nights later and added them to his catalog of objects that might confuse comet hunters.

M60 is similar in diameter to many spiral galaxies but, as an E2 elliptical, it is very nearly spherical, containing a much larger volume. It probably

has a mass of several trillion suns and is orbited by thousands of globular clusters. Using the Hubble Space Telescope to measure the motions of M60's stars, astronomers have discovered that a monster black hole of 4.5 billion solar masses lies at the galaxy's heart.

CLOSE NEIGHBORS

M60 lies very close to the spiral M59 (upper right), and the two galaxies are thought to be interacting. In a billion years, M60 may even swallow M59 entirely.



DISRUPTED SPIRAL GALAXIES

Antennae Galaxies



CATALOG NUMBERS
NGC 4038, NGC 4039

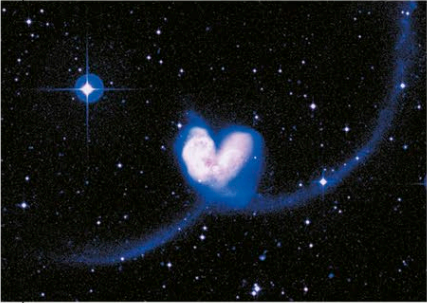
DISTANCE
63 million light-years

DIAMETER 360,000
light-years (total)

MAGNITUDE 10.5

CORVUS

The Antennae Galaxies, NGC 4038 and 4039, are among the sky's most spectacular interacting galaxies. Seen from Earth, they appear as a central bright double-knot of material, with two long streamers of stars stretching in opposite directions, resembling an insect's antennae. However, powerful telescopes reveal that each streamer is in fact a spiral arm, uncurled from its parent galaxy by the tremendous gravitational forces of an intergalactic collision that began around 700 million years ago and continues today.



THE BIGGER PICTURE

A wide-field view of the Antennae taken from Earth reveals both the bright, distorted cores and the long, faint streamers formed by the disrupted spiral arms.

The Antennae have been studied for what they can tell us about galaxy collisions. Detailed images of the central region show that it is lit by hundreds of bright, intense star clusters. These are thought to be forming as gas clouds in the galaxies become compressed by the collision, triggering starbursts (see the Cigar Galaxy, p.314). Astronomers can use the clusters' redness to estimate their age—older clusters emit redder light because the brighter blue stars are the most massive and therefore the first to die.



CLOUDS AND CLUSTERS

Turbulent dust clouds and brilliant star clusters appear in a Hubble view of the colliding Antennae Galaxies at right. The image above—a composite of a Hubble visible-light view with microwave observations from the Atacama Large Millimeter Array in Chile—reveals clouds of dense, cold gas (pink, red, and yellow areas) from which new stars are forming.

